ALESSANDRO CASTELLANI

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Portfolio: apmcastellani-projects.github.io · **LinkedIn**: Alessandro Castellani · **GitHub**: apmcastellani-projects

EDUCATION

Università Cattolica del Sacro Cuore — *Milan, Italy*

Master of Science (Laurea Magistrale) in Data Analytics for Business — *Sep 2025 – Expected 2027*

Coursework: Applied Linear Models, Algorithms & Deep Learning, Time Series, Computational Statistics, Optimization, Statistical Learning.

Università degli Studi dell'Insubria — *Como, Italy*

Bachelor of Science (Laurea Triennale) in Mathematics — *Graduated – Mar 2025*

Final Thesis: *"Reti Neurali per la Previsione di Performance Calcistiche"*. Modeled multi-season historical player metrics using TensorFlow/Keras with backpropagation and Adam optimization.

Pontificio Collegio Gallio — *Como, Italy*

Liceo Scientifico Internazionale (High School Diploma in Science) — *Sep 2016 – Jun 2020*

FEATURED QUANTITATIVE & ANALYTICS PROJECTS

Evaluating Offensive Production in Serie A (xG Linear Modeling) — *Academic Research (Cattolica) 2026*

Technologies: R, FBref/Opta Event Data, Multiple OLS Regression, Residual Diagnostics

- Constructed a multiple OLS regression model on 120 Serie A forwards analyzing xG generation beyond raw shot volume ($R^2 = 91.4\%$, Adj $R^2 = 90.8\%$); conducted tactical outlier diagnostics and validated stability out-of-sample.

IFCS 2026 Data Challenge — *Competitive Machine Learning (Milan-Bicocca) 2026*

Technologies: Python, Scikit-Learn, Imbalanced Classification, Pandas

- Engineered an ML classification pipeline predicting financial distress across 14,000 Italian SMEs under time constraints, pitching findings to an academic jury.
- Implemented non-linear signed logarithmic transformations ($\text{sign}(x) \cdot \ln(1 + |x|)$) for heavy-tailed balance sheets, 5-Fold Stratified CV, and Youden's J threshold optimization.

Multi-Horizon XGBoost Predictive Engine & Dashboard — *Time-Series ML Engineering 2026*

Technologies: Python, FastAPI, XGBoost, Purged Walk-Forward Cross-Validation, Plotly

- Built an end-to-end time-series return predictor utilizing purged walk-forward cross-validation with embargo buffers to eliminate lookahead bias; deployed REST API with interactive dashboard.

PROFESSIONAL EXPERIENCE

I Bravi SRL / SC Management SRL — *Various Locations, Italy*

Event & Festival Hospitality Staff — *Apr 2025 – Present*

- Delivered high-volume beverage service and guest hospitality at major music festivals (Nameless Festival, Snowland Festival; 20,000+ daily attendees), handling cash operations and POS reconciliation.

XROAD SSDRL — *Cadorago, Italy*

Operations & Front-Office Assistant (Part-Time) — *Jan 2025 – Feb 2026*

- Managed customer onboarding, CRM database records, and active subscription coordination for over 200 members; coordinated daily front-office administrative workflows.

Private Mathematics Tutoring — *Milan / Como, Italy*

Mathematics Tutor — *Sep 2020 – Present*

- Mentored 50+ secondary and university students in advanced calculus, linear algebra, and probability theory.

SOFT SKILLS

- Analytical Problem Solving, Team Coordination & Synergy, High-Pressure Execution, Client & Front-Office Relations.

TECHNICAL SKILLS & COMPETENCIES

- **Programming & Tools**: Python, R, SQL, LaTeX.
- **Statistical Learning**: Multiple OLS Regression, Residual Diagnostics & VIF, Time-Series Forecasting, Purged Walk-Forward CV, Non-linear Transforms.
- **Sports Analytics**: Opta & FBref Event Data, Expected Goals (xG) & xA, Role KPI Benchmarking, Scouting Radar Profiles.
- **Languages**: Italian (Native), English (Fluent / Professional Working Proficiency).